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Review article

Age at menopause is negatively associated with frailty: A systematic review and meta-analysis

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ABSTRACT

Menopause and related changes may be associated with frailty and contribute to higher frailty risk. This systematic review of the literature on the association between menopause and frailty combines the findings from studies of community-dwelling women. PubMed was systematically searched in March 2021 with a time frame from 2000 to March 2021 without language restriction. Potentially eligible studies were those that provided cross-sectional or prospective observational data on associations between menopause and frailty in community-dwelling women. Reference lists of relevant articles and the included studies were reviewed for additional studies. The same effect sizes were combined using a meta-analysis using the generic inverse variance method. From 131 studies identified, cross-sectional data on age at menopause from 3 studies and longitudinal data on surgical menopause from 2 studies were used for meta-analysis. Each one-year increase in age at menopause was significantly associated with a 2 % decreased risk of prevalent frailty (pooled odds ratio = 0.98, 95%CI (confidence interval) = 0.96–0.99, p < 0.001). Surgical menopause did not predict incident frailty (pooled OR = 1.02, 95%CI = 0.82–1.28, p = 0.23). This systematic review and meta-analysis showed that later age at menopause was significantly associated with a lower risk of prevalent frailty. In a clinical setting, age at menopause can be useful information to help clinicians to evaluate and stratify frailty risk in postmenopausal women. Hormonal changes after menopause may be related to the link between age at menopause and frailty and thus warrant further investigation.

1. Introduction

Natural menopause is defined as the permanent cessation of menstruation due to the loss of ovarian follicular function, and is confirmed by 12 consecutive months of amenorrhea with no other obvious pathological or physiological causes [1]. In most women, the onset of menopause ranges between ages 40 and 58, with the average age of onset 51 years [2]. Induced menopause can also occur at any time when a woman has surgical removal of both ovaries with or without hysterectomy or iatrogenic ablation of ovarian function by chemotherapy or radiation [1]. Increased risk of osteoporosis and fracture in postmenopausal women due to declining estrogen levels is well known, and the preventive measures have been developed and practiced [3]. However, recent studies have showed that risks of not only osteoporosis but also other diseases increase after menopause. For example, women who experience menopause earlier than usual, whether it is natural or induced, are at increased risk of cardiovascular diseases, neurological diseases, psychosexual dysfunction, mood disorders, and premature death, possibly due to estrogen deficiency [4]. Human life expectancy has dramatically improved in the last century [5]. While most of those who were born in 1900 did not survive until age of 50, the current average life expectancy for women is over 80 in many countries [5]. It means that a significant number of women experience a prolonged postmenopausal period of some 30 years, and even longer for those with earlier menopause. Nonetheless, women's health after menopause has been understudied by researchers and not properly addressed by healthcare providers, with scarce evidence in the literature [6].

Emerging from the field of geriatrics, frailty refers to a state

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associated with decreased physiological reserve and vulnerability to poor resolution of homeostasis when exposed to stress and resulting from age-related cumulative deficits in multiple body systems [7]. Frailty can be considered as a biological age, reflecting overall health status, and is known to be a strong predictor of mortality, independent of chronological age [7–9], as well as multiple negative health outcomes [10–14]. A sex disparity in frailty has been reported by numerous studies. Women have higher prevalence and greater severity of frailty than men at the same age, which is consistent in all age levels [15–17]. Although this fact is widely known and commonly accepted, the reason is still unknown. Various hypotheses were considered and previous studies explored biological, psychosocial, and behavioral factors as a cause of the sex disparity in frailty [18,19]. Among them, menopause and related changes may be associated with frailty and contribute to higher frailty risk in women. To date, only a few original studies have investigated the relationships between menopausal factors and frailty, showing conflicting results [20]. Therefore, the aim of this study is to conduct a systematic review of the literature for association between menopause and frailty and to combine the findings to synthesize pooled evidence.

2. Methods

The present study was conducted in accordance with the Preferred Reporting Items for Systematic Review and Meta-analyses (PRISMA) guideline [21].

2.1. Literature search

The PubMed database was systematically searched in March 2021 by one clinician researcher (GK) with a time frame between 2000 to March 2021, without language restriction, with explosion function if available. A protocol was developed a priori and was registered at PROSPERO (http://www.crd.york.ac.uk/PROSPERO/display_record.php?ID=CRD42021245690). The search terms used were ("Menopause" OR "Hysterectomy" OR "Oophorectomy") AND ("Frailty" OR "Frail elderly"). Titles, abstracts, and full texts were screened and examined for eligibility independently by two investigators (GK and YT). As other sources, reference lists of relevant articles and the included studies were reviewed for additional studies. Potentially eligible studies were ones that provided cross-sectional or prospective observational data on associations between menopause and frailty in community-dwelling women. Menopause-related factors of interest included age at menopause or reasons for menopause, such as surgical menopause. Frailty should be defined using validated criteria, such as the frailty phenotype or the Frailty Index. Randomized controlled trials, reviews, editorials, comments, conference abstracts, or dissertations were not considered. Corresponding authors were contacted if there was missing information.

2.2. Data extraction

The data extracted from included studies were first author, study cohort name, publication year, location, sample size, mean age and age range, frailty criteria used, mean age at menopause, follow-up period if the study was prospective, and findings.

2.3. Methodological quality assessment

The included prospective studies were assessed for methodological quality using the Newcastle-Ottawa scale for cohort studies, which includes nine items to cover three domains (selection, comparability, and outcome) [22]. A study which meets five items or more out of nine was considered to have adequate quality of methodology. As for the cross-sectional studies, methodological quality was assessed using the 8-item Joanna Briggs Institute Critical Appraisal Checklist for Analytical Cross-Sectional Studies [23]. A study meeting four items or more out of eight was considered to have adequate quality of methodology.

2.4. Statistical analysis

When two or more studies provided the same effect sizes regarding associations between menopause-related factors and frailty risk, a meta-analysis using the generic inverse variance method was conducted to combine these results and synthesize pooled effect sizes. The presence and degree of heterogeneity across the studies were examined using chi-square test and I^2 statistics, respectively. If significant heterogeneity was observed, a random-effects model was used, while a fixed-effects model was used when the significant heterogeneity was absent. Publication bias was assessed by visually inspecting funnel plots. All meta-analyses were conducted using the Review Manager 5 (version 5.2, The Cochrane Collaboration, Copenhagen, Denmark). $P < 0.05$ was considered to be statistically significant.

3. Results

3.1. Selection processes

The systematic literature search yielded 130 citations and 1 study was found through other source. Of 131 citations, 126 were excluded through screening of titles and abstracts, leaving 5 for full-text assessment. One study was further excluded due to not providing sufficient data, and a total of 4 studies were included in this review (Fig. 1). An updated PubMed search on 21 June 2022 during 2021–2022 did not identify any new publications.

3.2. Study characteristics

The characteristics of included studies and the participants are summarized in Table 1. The studies were from the UK [24], the US [25], South Korea [26], and Canada [27]. Mean age at menopause ranged from 47.9 [24] to 49.9 [26] years old. The follow-up time of two longitudinal studies was 4 years [24] and 18 years [25]. Two studies showed frailty prevalence of 11.6 % [24] and 11.2 % [26]. Two studies [24,25] longitudinally examined risk of incident frailty and the remaining two studies [26,27] examined cross-sectional associations between menopause and prevalent frailty. The sample sizes ranged from 1249 [24] to 15,320 [27]. CHS criteria was used by three studies [24,26,27], while the Study of Osteoporotic Fractures criteria [25] and the Frailty Index [27] were used once. Data on associations between prevalence of frailty and age at menopause were provided by three studies [24,26,27]. Data on incident frailty risks by age at menopause [24] and surgical menopause [24,25] were provided by two longitudinal studies. The definitions of surgical menopause were having both ovaries removed before menopause [25] or answering "surgery" as a cause of menopause at interview [24]. Only one study provided prevalence and duration of hormone replacement therapy (HRT) and showed that frail women were significantly less likely to use HRT compared (7.1 %) with non-frail women (19.1 %) [26].

Two longitudinal studies were considered to have adequate methodological quality based on the Newcastle-Ottawa scale (Table 2: number of studies = 2, mean = 6.5, range = 6–7). Three studies providing cross-sectional associations between age at menopause and prevalent frailty were considered to have adequate methodological quality based on the Joanna Briggs Institute Critical Appraisal Checklist for Analytical Cross-Sectional Studies (Table 3: number of studies = 3, mean = 8, range = 8).

3.3. Meta-analysis of age at menopause and prevalent frailty

OR (odds ratios) of prevalent frailty risk per one-year increase in age at menopause were available from three studies [24,26,27]. All effect measures were adjusted, and covariates used by two or more studies

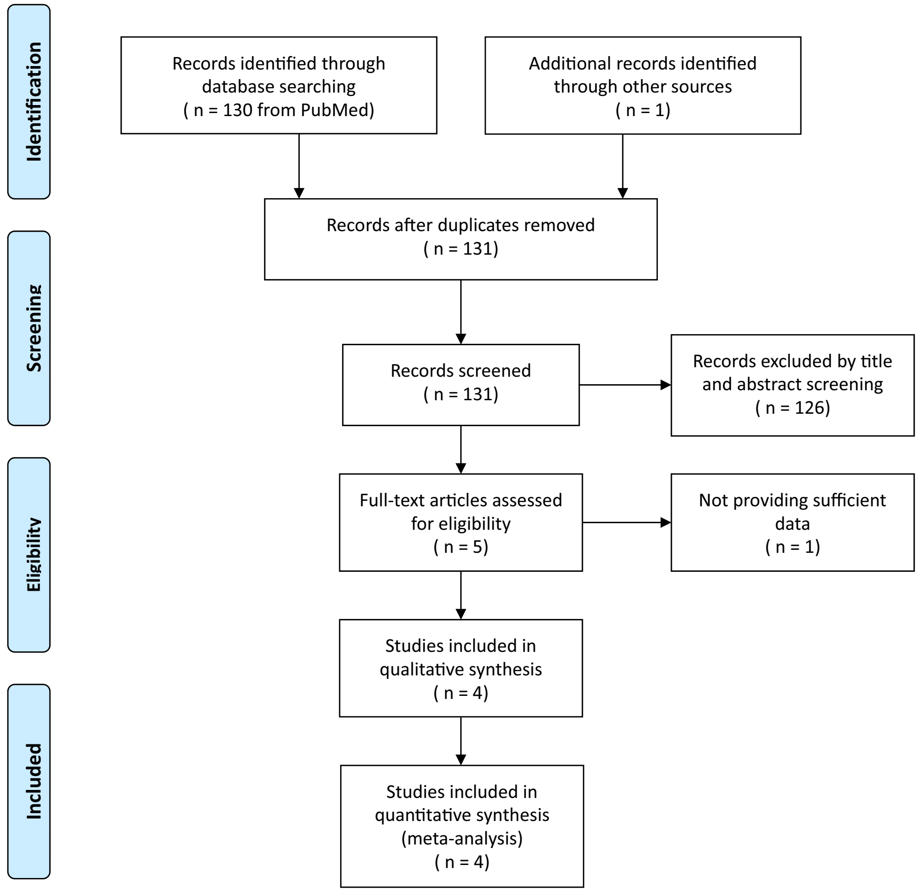

Fig. 1. Flow chart of systematic literature review.

were age, education, marital status, smoking, alcohol, income, and hormone replacement therapy use. A fixed-effects meta-analysis was used to combine the effect sizes as the heterogeneity was low ($I^2 = 12$ %, $p = 0.32$). Each one year increase in age at menopause was significantly associated with a 2 % decreased risk of prevalent frailty (pooled odds ratio = 0.98, 95%CI (confidence interval) = 0.96–0.99, $p < 0.001$) (Fig. 2).

3.4. Meta-analysis of surgical menopause and incident frailty

Two studies examined the risk of incident frailty in women who had surgical menopause compared with that in women who had natural menopause. Both studies used various covariates for adjustment. The surgical menopause did not predict incident frailty (pooled OR = 1.02, 95%CI = 0.82–1.28, $p = 0.23$) (Fig. 3).

Publication bias assessment, subgroup analysis, and sensitivity analysis were not able to be conducted due to the limited number of studies included for abovementioned meta-analyses.

4. Discussion

This study has conducted a systematic literature search for studies examining associations between menopausal factors and frailty risk in community-dwelling women. The meta-analysis showed that each 1-year increase in age at menopause was significantly associated with a 2 % decreased risk of prevalent frailty, while surgical menopause did not predict development of frailty.

Although specific underlying mechanisms for the association between age at menopause and frailty have not been established, hormonal changes following menopause, especially estrogen deficiency, are considered to play a role. Hormonal replacement can mitigate not all but some of the negative consequences of earlier menopause, such as cardiovascular diseases, neurological diseases, and premature death [4,28]. In fact, hormonal replacement therapy is recommended by international authorities for women with premature menopause or premature ovarian failure [4]. Estrogens, one of the important sex hormones, are mostly produced in the ovaries and released into the circulation, and affect various target tissues [29]. A steep decline of circulating estrogen levels occurs at menopause and may be the potential cause of different frailty risks according to age of menopause onset. One possible explanation for the association between earlier menopause and higher frailty risk is loss of estrogen's protective effects on muscle. Estrogen receptors are found in the nuclei of human muscle fibers and capillaries, and some evidence suggested that estrogen preserves muscle mass, strength, and functions [30,31]. Women with earlier onset of menopause may be more predisposed by the decreased estrogen levels to these negative conditions and, therefore, to the development of frailty. Another possibility is inflammation. Estrogen plays an important role in immune and inflammatory processes [29]. Emerging evidence showed that a decline in estrogen is closely associated with systemic chronic inflammation and increased risks of cancer, cardiovascular disease, neurodegeneration, and stroke [29,32]. These factors may facilitate the development of frailty in postmenopausal women.

Women who had surgical menopause may as well be predisposed to frailty. In addition, reasons for surgical menopause, such as malignancies, and burden of surgery may add further risks. However, surgical menopause did not increase incident frailty risk according to the current meta-analysis. Counterintuitive findings could be a result of the limited number of studies ($n = 2$) or hormone replacement therapy initiated after surgical menopause. Further studies with appropriate adjustments for important related factors, such as hormonal replacement therapy or reasons for surgery, are warranted.

Table 1
Summary of included studies on associations between menopause and frailty risk.

Author/ year/study	Location	Sample size	Mean age (range)	Frailty criteria (prevalence)	Mean age at menopause	Follow-up if prospective	Findings
Kojima 2022 ELSA	UK	1249	- (≥60)	mCHS (11.6 %)	47.9	4 years	Logistic regression models for incident frailty adjusted for age, smoking, alcohol, wealth, education, marital status, age at menarche, and number of children. - aOR = 0.97 (0.95–1.00) for one increase in age at menopause - aOR = 2.00 (1.35–2.97) and 1.08 (0.52–2.24) for women with menopause at 10–45 and 56–75 years, respectively, compared with menopause at 46–55. - aOR = 1.27 (0.82–1.97) for surgical menopause compared with natural menopause.
Huang 2018 SOF	US	7699	71.2 (≥65)	SOF (–)	48.2	18 years	Multinomial logistic regression models for incident frailty adjusted for age, BMI, and IADL disabilities. - aOR = 0.94 (0.72–1.22) for surgical menopause compared with natural menopause
Lee 2020 KFACS	South Korea	1264	75.6 (70–84)	mCHS (11.2 %)	49.9	–	Logistic regression models for prevalent frailty adjusted for age, marital status, education, diabetes, hypertension, polypharmacy, hospitalization, falls and hormone replacement therapy. - aOR = 0.95 (0.91–0.98) for one increase in age at menopause - aOR = 0.83 (0.32–2.15), 1.11(0.24–1.41), and 0.89 (0.13–1.06) for menopause at 40–45, 46–54, and 55–64 years, respectively, compared with menopause at 30–39.
Verschoor 2019 CLSA	Canada	15,320	- (45–85)	mCHS (–) FI	–	–	Logistic regression models for prevalent frailty defined by the frailty phenotype adjusted for age, marital status, ethnicity, coresidence, smoking, alcohol, income, education, social support, and hormone replacement therapy use ever. - aOR = 0.98 (0.96–1.01) for one increase in age at menopause - aOR = 1.33 (0.72–2.27), 1.11(0.80–1.51), and 0.89 (0.64–1.23) for menopause at 30–40, 41–45, and 55+ years, respectively, compared with menopause at 46–55. Linear regression models for FI adjusted for age, marital status, ethnicity, coresidence, smoking, alcohol, income, education, social support, and hormone replacement therapy use ever. - β = −0.0012 (−0.0015 to −0.0009) for one increase in age at menopause - β = −0.024 (−0.016 to −0.032), 0.007 (0.002–0.013), and −0.004 (−0.01 to 0.001) for menopause at 30–40, 41–45, and 55+ years, respectively, compared with menopause at 46–55.

aOR: Adjusted odds ratio.
BMI: Body mass index.
CLSA: Canadian Longitudinal Study of Aging.
ELSA: English Longitudinal Study of Ageing.
FI: Frailty Index.
IADL: Instrumental activities of daily living.
KFACS: Korean Frailty and Aging Cohort Study.
mCHS: Modified Cardiovascular Health Study criteria.
SOF: Study of Osteoporotic Fractures criteria.

Table 2
Methodological quality assessment with the Newcastle-Ottawa Quality Assessment Scale for cohort studies.

Author/ year	Representativeness	Non-exposed group	Ascertainment of exposure	Absence of outcome at baseline	Controlling for important factors	Controlling for additional factors	Ascertainment of outcome	Length of follow-up	Adequacy of follow-up	total
Kojima 2021	1	1	0	1	1	1	1	1	0	7/9
Huang 2018	1	1	0	1	1	0	1	1	0	6/9

Table 3
Methodological quality assessment with the Joanna Briggs Institute Critical Appraisal Checklist for Analytical Cross-Sectional Studies.

Author/year	Inclusion	Setting	Exposure	Measurement	Confounders	Strategies for confounders	Outcome	Statistical analysis	total
Kojima 2021	1	1	1	1	1	1	1	1	8/8
Lee 2020	1	1	1	1	1	1	1	1	8/8
Verschoor 2019	1	1	1	1	1	1	1	1	8/8

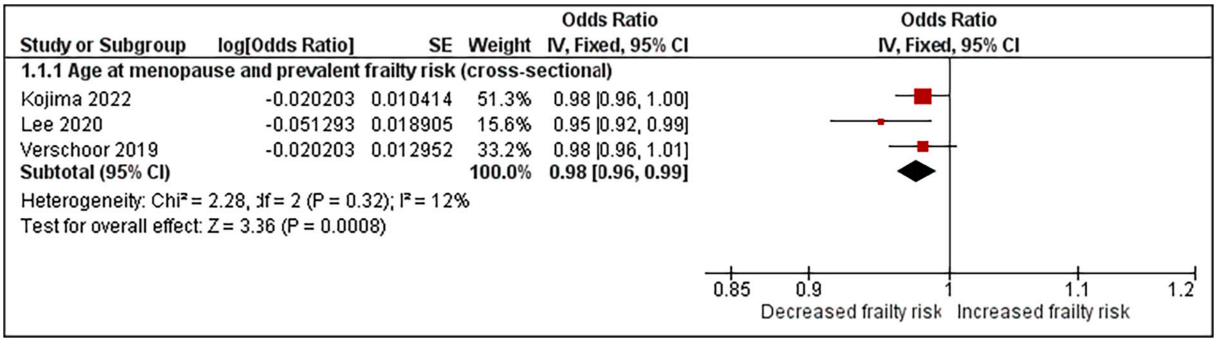

Fig. 2. Forest plot of prevalent frailty risk according to one increase in age at menopause.

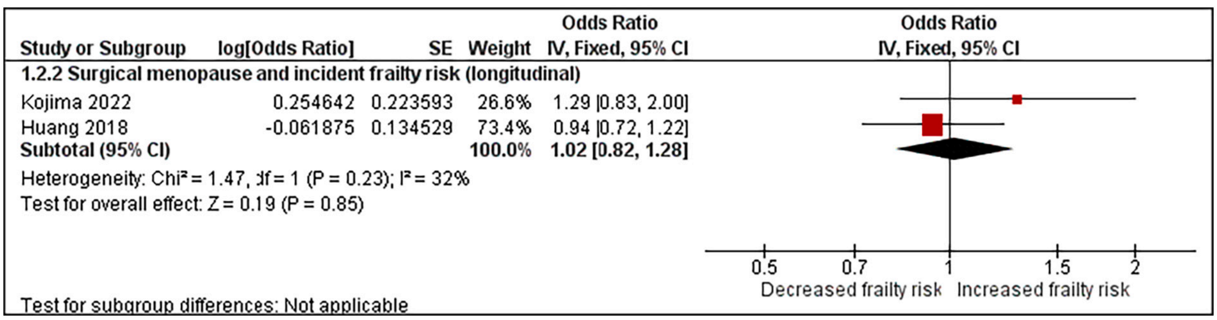

Fig. 3. Forest plot of incident frailty risk according to surgical menopause compared with natural menopause.

To the best of our knowledge, this is the first systematic review and meta-analysis focusing on associations between menopausal factors and frailty risk, and successfully providing significant pooled evidence that women with later menopause have lower risk of frailty. The strengths of this study include the robust methodology following the PRISMA statements. The search strategy was comprehensive and reproducible, using the MeSH and text terms, and free of language restriction. The screening of title, abstract, and full-text was done independently by two researchers. In addition, meta-analysis included adequately adjusted effect measures from the included original studies. However, our study is not without limitations. First, the number of the included studies was relatively small, perhaps because reproductive factors are still an understudied area in the frailty research. Nonetheless, it is noteworthy that all three studies included in the meta-analysis on age at menopause and prevalent frailty provided coherent results with low heterogeneity and yielded significant pooled evidence. Second, the limited number of included studies restricted the use of publication bias assessment and subgroup analysis, and sensitivity analysis. Third, only one database was used for literature search and some important articles may have been missed. Fourth, although women who had surgical menopause before natural one could be a confounding factor in the relationship between age at menopause and frailty risk, it was not taken into consideration in the analyses of included studies. Lastly, only cross-sectional data on the association between age at menopause and frailty risk were available for meta-analysis, therefore, a causal relationship could not be inferred. There is only one prospective study found in the literature, which showed that earlier menopause was a significant predictor of frailty [24].

5. Conclusion

In conclusion, later age at menopause was significantly associated with lower risk of prevalent frailty in community-dwelling women. Hormonal changes after menopause may be related to the link between age at menopause and frailty and thus warrant further investigation. These findings enhance our understanding of pathophysiology of frailty development and, possibly, sex disparity of frailty.

Contributors

Gotaro Kojima participated in conception and design of the study, acquisition of data, analysis and interpretation of data, drafting, and revising critically for important intellectual content.

Yu Taniguchi participated in conception and design of the study, acquisition of data, analysis and interpretation of data, and revising critically for important intellectual content.

Kohei Ogawa participated in analysis and interpretation of data, and revising critically for important intellectual content.

Reijiro Aoyama participated in conception and design of the study, acquisition of data, analysis and interpretation of data, and revising critically for important intellectual content.

Tomohiko Urano participated in analysis and interpretation of data, and revising critically for important intellectual content.

All authors saw and approved the final version and no other person made a substantial contribution to the paper.

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Declaration of competing interest

The authors declare that they have no competing interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.maturitas.2022.07.012.

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